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# Hallucination Is Relative: Evaluating LLM Divergence Under Truth-Regime Contracts

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## Abstract

Large language models sometimes state things that are not true, but whether that matters depends on the task. Inventing a diagnosis is dangerous; inventing a metaphor may be exactly what the user asked for. Yet current benchmarks treat hallucination as a binary defect, ignoring what the prompt actually authorizes. We propose evaluating hallucination *relative to the prompt’s truth regime*: what must be true, what may be uncertain, and what may be invented. Our study (12,469 texts, 10 models, 310K claims) shows that creative intent raises hallucination (+12.8 %), the requested writing format has the strongest effect (+21.4 %), and the factual constraints expected by each writing format reduce how much creativity raises hallucination, supporting the view that hallucination is regime-relative, not absolute.

## 1 INTRODUCTION: THE CREATIVITY-HALLUCINATION PARADOX

Large language models (LLMs) sometimes state things that are not true. The community calls these errors *hallucinations*: the model produces fluent, confident text that is not supported by evidence or contradicts known facts [Ji et al., 2023, Huang et al., 2025]. Hallucination is widely regarded as the central reliability barrier for deploying LLMs in high-stakes settings such as medicine, law, and education [Lin et al., 2022]. Consequently, a large body of work aims to detect, measure, and reduce hallucination [Li et al., 2023, Wei et al., 2024, Thorne et al., 2018, Chen and Eger, 2023].

Yet there is a paradox. When users ask an LLM to write fiction, brainstorm, or propose creative solutions, they actively *want* the model to go beyond established facts. In these

settings, punishing every unsupported statement would suppress the very outputs users seek [Franceschelli and Musolesi, 2024, Hou et al., 2025, Rick et al., 2023]. A brainstorming tool that only mentions ideas that already exist is useless.; a medical assistant that invents diagnoses is dangerous. The difference is not in the text itself, but in *what the prompt asked for*.

**The diagnostic gap.** Current evaluation frameworks do not account for this difference. On one side, hallucination benchmarks treat any unsupported content as a defect and seek to minimize it [Bang et al., 2025]. On the other side, creativity benchmarks treat novelty as a virtue and seek to maximize it, often without checking factual grounding [Hou et al., 2025, Franceschelli and Musolesi, 2024, Acar et al., 2019]. Neither tradition conditions its scores on what the user’s prompt actually authorizes. As a result, the same output can be scored as a serious failure by one framework and a creative success by another, depending on which evaluation protocol is applied, not on whether the divergence was appropriate given the user’s intent. We examine both traditions and their shared blind spot in Section 3.

Recent theoretical work shows that LLMs cannot avoid hallucination entirely: formal proofs via incompleteness and undecidability [Banerjee et al., 2024] and calibration arguments [Kalai et al.] establish a nonzero *structural* error floor. Yet the same mechanism also produces novel hypotheses and creative combinations that users request in open-ended tasks [Sun et al., 2024, He et al., 2025a, Sui et al., 2024]. In Section 3, we argue that the goal should be to *control* divergence relative to the task’s *truth regime* (TR), not to eliminate it.

**Our thesis.** We claim that hallucination is not a binary property of text. It is *relative to the prompt’s truth regime* (TR): the task-specific contract that specifies what must be true, what may be uncertain, and what may be invented. Under a strict TR (e.g., “summarize this document accurately”), any unsupported claim is a hallucination (H). Under

a permissive TR (e.g., “write a fiction”), that same divergence may instead constitute *productive divergence* (PD): a regime-compliant departure from established facts that is novel and task-useful, and therefore fulfills rather than violates the user’s creative intent (CI). The distinction is not about the text itself, but it is about whether the prompt authorized the deviation. Evaluation that ignores this conflates H with PD.

**Empirical motivation.** We ground this claim empirically with 12,469 texts generated by 10 LLMs, across prompts varying task type, stylistic creativity, and output format under a strict TR (no factual content invention); each text was decomposed into atomic claims and checked against Wikipedia as a factual knowledge base [Bang et al., 2025]. Three findings follow: (i) higher CI raises hallucination by 12.8 %; (ii) writing format is the largest single driver (+21 %); and (iii) format-specific factual constraints moderate how much creativity raises hallucination, supporting the view that hallucination is TR-relative, not absolute. **This paper makes three contributions.** *First* (conceptual), we introduce a minimal vocabulary (TR, CI, PD, H) that makes the prompt contract explicit (Section 2). *Second* (analytical), we survey hallucination and creativity evaluation, show their shared blind spot, and argue that universal suppression of H is infeasible given H’s structural floor (Section 3). *Third* (empirical), a large-scale experiment (12,469 texts, 10 models) shows that the creativity and hallucination link is real, task-dependent, and TR-relative (Section 4); we then outline a six-point research agenda (Section 5).

## 2 TERMINOLOGY AND CONCEPTUAL FRAMEWORK

**Motivation.** The literature uses the word *hallucination* broadly, but not consistently. Surveys commonly group together factuality errors against world knowledge and faithfulness errors against a provided source, sometimes adding axes such as intrinsic vs. extrinsic causes and different granularities Ji et al. [2023], Huang et al. [2025], Zhang et al.. Benchmarking work further argues that hallucination is often conflated with factuality and that evaluation depends on the chosen oracle and evaluator reliability Bang et al. [2025]. To study creativity-hallucination tradeoffs, we fix a minimal vocabulary that makes the *prompt contract* explicit. Throughout, we distinguish *prompt-level* variables (what the user requests) from *output-level* properties (what the model produces), and adopt a *claim-level* unit of analysis because evaluations explicitly vary by granularity.

**Creative intent (CI; prompt-level).** The extent to which the user *requests novelty* in the response (e.g., alternatives, analogies, imaginative options) *under stated constraints*. CI increases the preferred exploration of the solution space; creativity is usually defined as *novel and useful/appropriate*

relative to given constraints. Runco and Jaeger [2012], Acar et al. [2019], Franceschelli and Musolesi [2024].

**Truth regime (TR; prompt-level contract).** A task-specific contract that determines (i) what must be *grounded/true*, (ii) what may be *uncertain* (but should be labeled), and (iii) what may be *invented*, together with the relevant *oracle* (source document, retrieval set, database, world facts). TR is necessary because “hallucination” definitions and evaluation oracles vary across settings Bang et al. [2025]. Critically, we separate two independent permissions within a regime: *stylistic latitude* ( $\sigma$ ), how freely form may vary while preserving propositions Jin et al. [2022] and *factual content invention* ( $\kappa$ ), whether new factual content may be introduced. Many real prompts request creative style without factual content invention (e.g., “explain photosynthesis like a poet”); here  $\sigma$  is high but  $\kappa$  remains low, so figurative language is welcome but fabricated claims are not Franceschelli and Musolesi [2024].

**Faithfulness (F; output-level).** Whether the output *adheres to the prompt* and is *supported by the provided evidence* (documents, retrieved passages, tables), without introducing unsupported details. This is the central failure mode in grounded generation and summarization, where fluent outputs can exceed source support Maynez et al. [2020], Kryscinski et al. [2020], Huang et al. [2025], Malin et al. [2025].

**Accuracy / factuality (A; output-level).** Whether the claims that must be true under TR are actually supported by an appropriate external source (e.g., references, gold labels, or world knowledge). This corresponds to QA and fact-checking evaluation.

**Hallucination (H; claim-level).** As we define, a hallucination is a TR-violation. The model presents a claim *as grounded/true* where TR requires truth or evidence, but the claim is unsupported (F) and/or incorrect (A). This definition unifies faithfulness and factuality failures while making the oracle explicit Maynez et al. [2020], Bang et al. [2025].

**Productive divergence (PD; claim-level).** Content that would count as hallucination under a strict factual standard, but is allowed by the prompt’s truth regime and helps accomplish the task. Formally, PD is a TR-compliant exploration that proposes *novel* candidates (hypotheses, alternatives, analogies, fictional details) where TR permits speculation or invention; it is *productive* when the deviation is useful and appropriate under task constraints—the standard creativity criterion Diedrich et al. [2015], Acar et al. [2019]. This framing echoes non-LLM uses of “productive divergence” as variation that energizes performance Parry et al. [2023], Lingard et al. [2017]. Crucially, if TR demands grounding, the same deviation is H, not PD.

**Decision rule (simple and testable).** Given prompt  $p$  and output  $y$  and an oracle  $O$ : specify  $\text{TR}(p)$ , decompose  $y$

into atomic claims, then label each claim using  $O$  as H (truth-conditional under TR but unsupported/incorrect), PD (licensed by TR and task-useful), or neither (stylistic variation, benign noise), see Appendix B.1 for the step-by-step procedure.

Under this vocabulary, we do not oppose “creativity vs. truth”. We oppose **H** (TR violations) to **PD** (TR-compliant creativity). Intent-agnostic factuality evaluation fails to separate these two outcomes.

## 2.1 TRUTH REGIMES AND REGIME-RELATIVE DIVERGENCE

The terms above give us a vocabulary for individual claims; we still need a way to specify *what the task allows*. We formalize this as  $\text{TR}(p) = (O, \sigma, \kappa)$ : an oracle  $O$  (the reference against which claims are checked), a stylistic latitude  $\sigma$ , and an invention license  $\kappa$  Ji et al. [2023], Huang et al. [2025]. Table 1 instantiates three canonical regimes. Existing benchmarks typically fix  $O$  but rarely annotate  $(\sigma, \kappa)$  explicitly, making it easy to conflate PD with H Bang et al. [2025], Jiang et al. [2024].

**Claim-level labeling.** Given a regime  $\text{TR}(p)$ , each atomic claim falls into one of the categories already defined: *stylistic variation* (form changes, no new propositions), PD (novel content licensed by  $\kappa$  and task-useful), H (truth-conditional content unsupported by oracle  $O$ ), or *benign noise* (minor inaccuracies with negligible impact) Ji et al. [2023], Huang et al. [2025], Bang et al. [2025].

**Two-sided failure relative to  $\kappa$ .** When  $\kappa$  is low, excess deviation increases H; when  $\kappa$  is high, *insufficient* deviation yields conservative outputs that fail the creative objective, a failure mode captured by creativity benchmarks but not by hallucination-only scoring Hou et al. [2025]. Our study varies *only*  $\sigma$  (style) while keeping  $\kappa$  minimal; therefore, any increase in factual deviation remains H under the tested regime.

With this vocabulary and labeling procedure in place, the natural question is: do existing evaluation frameworks already capture the TR/CI distinction? The next section examines both hallucination and creativity benchmarks and shows that they do not.

## 3 EVALUATION LANDSCAPE AND THE CASE FOR CONTROLLED DIVERGENCE

**Two traditions, one shared blind spot.** Hallucination benchmarks treat divergence from an oracle as a defect to minimize [Ji et al., 2023, Huang et al., 2025, Bang et al., 2025]; creativity benchmarks treat it as evidence of novelty to maximize [Hou et al., 2025]. This division holds for

single-intent tasks but breaks down for mixed-intent prompts that combine novelty with factual constraints [Wang et al., 2024b]: the same surface deviation can be PD or H, yet on both sides the prompt contract  $(O, \sigma, \kappa)$  remains implicit. Hallucination pipelines assume  $\kappa$  is low; creativity pipelines score novelty without specifying which claims are bound to an oracle, making it easy to conflate regime-compliant exploration with genuine TR-violations.

### 3.1 HALLUCINATION EVALUATION: FACTUALITY, FAITHFULNESS, AND REASONING VALIDITY

Hallucination evaluation is anchored in the choice of an oracle, either *world knowledge* (A: factuality) or *provided context* (F: faithfulness) [Ji et al., 2023, Huang et al., 2025], with additional axes such as contradiction vs. fabrication, intrinsic vs. extrinsic errors [Zhang et al.], and evaluation granularity (token/span/claim/output) [Bang et al., 2025]. Accordingly, modern protocols increasingly rely on claim-level decomposition and automated or semi-automated verification, e.g., retrieval+Natural Language Inference (NLI) [Chen and Eger, 2023, Hughes et al., 2023] or QA-based checking [Wang et al., 2020], building on fact-checking [Thorne et al., 2018] and truthfulness benchmarks [Lin et al., 2022, Li et al., 2023]. A key methodological motivation is diagnosticity: explicit claims make oracle comparison and error typing more interpretable than output-level judgments [Bang et al., 2025].

However, these axes are necessary but not sufficient for TR/CI-aware judgment. Most hallucination pipelines implicitly assume a strict invention license ( $\kappa$  low): any unsupported truth-conditional content is treated as a failure. In grounded generation, this assumption is defensible, but it collapses when prompts license speculation or invention. Even in summarization, a statement can be factually correct (A satisfied) yet unsupported by the source (F violated), showing that H depends on the oracle and TR, not on the text alone [Maynez et al., 2020, Kryscinski et al., 2020]. Without explicit  $(O, \sigma, \kappa)$  metadata, the same verification pipeline can conflate stylistic variation, factual content invention, and regime violations.

### 3.2 CREATIVITY EVALUATION: FRAGMENTED METRICS AND EVALUATOR DEPENDENCE

Creativity is commonly defined as combining *novelty* with *value/appropriateness* under constraints [Runco and Jaeger, 2012], but operationalizations and scoring procedures vary sharply across tasks and domains [Acar et al., 2019, Diedrich et al., 2015, Franceschelli and Musolesi, 2024, Ismayilzada et al., 2025a]. Recent LLM work therefore treats creativity as multi-dimensional and task-dependent, with benchmarks spanning (i) divergent-thinking tests and

Table 1: **Canonical truth regimes, specified by an oracle  $O$  and prompt-level permissions.** Here,  $\sigma$  denotes stylistic latitude (how freely form can vary), and  $\kappa$  denotes the factual content invention (whether new truth-conditional content may be introduced). The same surface “deviation” can be acceptable or a violation depending on whether the regime binds claims to provided evidence/world facts (faithfulness/factuality) or instead licenses invention under other constraints Maynez et al. [2020], Kryscinski et al. [2020], Huang et al. [2025], Franceschelli and Musolesi [2024].

| Regime                                            | Oracle $O$                                                   | Permissions<br>( $\sigma, \kappa$ ) | Allowed deviation                                                 | TR-violation (hallucination trigger)                                                           |
|---------------------------------------------------|--------------------------------------------------------------|-------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Strict grounded generation / summarization</b> | Provided sources (docs / retrieved passages)                 | (low, low)                          | Paraphrase, compression, re-ordering; <i>no new facts</i> .       | Any unsupported detail, added entity/number, or contradiction vs. sources.                     |
| <b>Factual task with creative style</b>           | World knowledge and/or cited sources                         | (high, low)                         | Metaphor/voice/vividness <i>without changing propositions</i> .   | Any new truth-conditional claim without support or incorrect claim.                            |
| <b>Ideation / fiction-writing</b>                 | Prompt constraints (outline, coherence, user-provided facts) | (high, high)                        | Invent entities, scenes, alternatives; explore hypotheses freely. | Violating explicit constraints (e.g., incoherence; contradicting user-provided facts/outline). |

Torrance Tests of Creative Thinking (TTCT) like rubrics [Organisciak et al., 2023, Nguyen and Singla, 2025, Dinu and Florescu, 2024, Zhao et al., He et al., 2025b], (ii) creative-writing evaluation (rubric- and reference-based) [Fein et al., 2025, Ippolito et al., 2022, Ismayilzada et al., 2025b, Li et al., 2024, 2025a, DeLorenzo et al., 2024, Wang et al., 2024a], and (iii) constrained creativity in code/math and interactive problem solving [Lu et al., 2025b, Ye et al., 2024, Qian et al., 2025, Tian et al., 2025, Hou et al., 2025]. CREATIVITYPRISM exemplifies this trend by aligning heterogeneous tasks to shared dimensions (quality, novelty, diversity) [Hou et al., 2025] while retaining task-local scoring rules (rubric-based criteria, dispersion/diversity signals, and constrained correctness–novelty trade-offs) [Lu et al., 2025b, Ye et al., 2024, Lu et al., 2025a].

Creativity judgments also remain evaluator-dependent: ratings shift with rubric design [Shively et al., 2018, Dietz et al., 2025, Marco et al., 2025], rater backgrounds [Cyriac, 2025, Ismayilzada et al., 2025b, Jr et al., 2025], and reader profiles [Ippolito et al., 2022], changing model rankings even under identical prompts when using LLM-as-judge protocols [Liu et al., 2023, Zheng et al., 2023, Li et al., 2025b, Han et al., 2025].

From a TR/CI perspective, however, the deeper issue is that creativity benchmarks rarely treat the truth regime as an explicit variable. Novelty signals may be rewarded without distinguishing PD from H [Sui et al., 2024, Padmakumar et al., 2025], even when factual grounding is part of the prompt contract [Wang et al., 2024b, Bang et al., 2025]. Existing faithfulness metrics [Maynez et al., 2020, Pagnoni et al., 2021, Kryscinski et al., 2020] and semantic verification tools [Wang et al., 2020, Aynedinov and Akbik, 2024, Chen and Eger, 2023] could detect such violations, but they are not integrated into creativity evaluation pipelines. This omission, shared with hallucination benchmarks, produces a concrete failure mode.

### 3.3 FAILURE MODE: CI-AGNOSTIC EVALUATION CONFLATES PD WITH H

When TR is left implicit, two symmetric errors arise: creativity benchmarks [Hou et al., 2025, Fein et al., 2025] assume multiple valid outputs without conditioning on an external oracle, while hallucination benchmarks [Bang et al., 2025, Lin et al., 2022] treat unsupported claims as errors relative to a specified oracle. Under a permissive regime ( $\kappa$  high), strict factuality checks or semantic verification misclassify licensed speculation as H (false positives on PD), an issue documented for both human [Cyriac, 2025] and automated raters [Sui et al., 2024]. Under a strict regime ( $\kappa$  low), novelty-oriented scores [Liu et al., 2023, Zheng et al., 2023] reward confident fabrications that boost apparent originality (false negatives on H) [Chen and Eger, 2023, Dietz et al., 2025, Li et al., 2025b]. Even benchmarks with domain-specific correctness constraints [Lu et al., 2025b, Ye et al., 2024, Hou et al., 2025, DeLorenzo et al., 2024] underspecify the prompt-level contract as a general variable, motivating a minimal evaluation layer that makes  $(O, \sigma, \kappa)$  explicit [Wang et al., 2024b, Boubdir et al.].

Beyond evaluation gaps, there is a deeper reason why the PD/H confusion persists: hallucination cannot simply be engineered away.

### 3.4 WHY UNIVERSAL SUPPRESSION OF H IS NEITHER FEASIBLE NOR DESIRABLE

LLMs are probabilistic next-token predictors trained under incomplete supervision; they produce plausible continuations even when the truth is underspecified, so eliminating H universally would require restricting the model or the task class. Formal arguments support this claim [Banerjee et al., 2024, Cossio, 2025], and empirical evidence confirms that even strict TRs reveal large residual error rates on short-

form QA [Mallen et al., 2023, Cheng et al., 2023, Gruber et al., 2025], truthfulness benchmarks [Lin et al., 2022, Wei et al., 2024], and domain-specific evaluations [Rahman et al., 2024, Sadat and Caragea, 2024, Uluoglakci and Temizel, 2024]. HalluLens confirms that H and A are distinct constructs, making the relevant floor inherently TR-dependent [Bang et al., 2025].

Yet the same generative mechanism that produces unsupported claims also produces the novel hypotheses and creative combinations that users request in open-ended tasks. Several systems exploit this deliberately through a *diverge-then-converge* pattern: first induce diverse candidates, then filter or verify to meet the task’s TR. Sun et al. [2024] formalize “good hallucinations” as reasoning paths that remain correct after convergence; Banerjee et al. [2025] show that verification-style mitigation can preserve divergent thinking whereas certain decoding interventions suppress it. Concrete instantiations, including creative problem-solving scaffolds [Rick et al., 2023], intent-hypothesis grounding [Bodonhelyi et al., 2024], logic-to-language transfer [You et al., 2025], and style and correct fine-tuning [Roychowdhury et al., 2024]. All treat divergence as a resource controlled by a subsequent convergence step, not a defect.

Since H has a structural floor and the same divergence drives PD, the practical goal cannot be universal suppression of H but TR-calibrated control: specify the regime upfront, enable divergence where CI warrants it, and enforce convergence before any output is presented as fact. This perspective aligns with evidence that divergence improves outputs valued in creative contexts (PD under permissive TR) yet becomes H under strict regimes [Sui et al., 2024, He et al., 2025a]. Operationally, such control can take the form of explicit mode signaling (hypothesis vs. claim), selective abstention, multi-sample generation with verification, or human-in-the-loop selection when automated judges are brittle [Bang et al., 2025, Li et al., 2025b].

## 4 OUR STUDY: A CONTROLLED EXPERIMENT AND EMPIRICAL FINDINGS

The preceding sections argue that H is TR-relative and that creative prompting alone, without granting factual content invention ( $\kappa$  low) should measurably raise deviation under a strict regime. No existing benchmark tests this claim directly across tasks and creativity levels. This section addresses that gap: does increasing stylistic CI while holding  $\kappa$  low actually increase claim-level H, and by how much? The verification pipeline follows HalluLens [Bang et al., 2025], which advocates separating H from A and evaluating at the claim level for diagnosticity. Concretely, all three task types share the same TR: the prompt requests content grounded in Wikipedia knowledge ( $O$  = world knowledge),

Table 2: Main effects on hallucination rate ( $N = 12,469$ ; 10 models, 53 runs). Per-condition mean,  $\Delta\%$  vs. ref. ( $^\dagger$ ), Spearman  $\rho$ , FDR-corrected  $p$ . Interaction effects in text.

| Factor | Cond.                | Mean  | $\Delta\%$ | $\rho$ | $p$           |
|--------|----------------------|-------|------------|--------|---------------|
| Task   | INTERV. <sup>†</sup> | 51.3% | —          | .450   | < $10^{-300}$ |
|        | NEWS                 | 56.4% | +5.2       |        |               |
|        | LESSON               | 72.6% | +21.4      |        |               |
| Creat. | FACTUAL <sup>†</sup> | 54.7% | —          | .295   | < $10^{-248}$ |
|        | HYBRID               | 60.5% | +5.8       |        |               |
|        | V.CREAT.             | 67.5% | +12.8      |        |               |
| Length | 250 w. <sup>†</sup>  | 57.8% | —          | .106   | < $10^{-32}$  |
|        | 500 w.               | 59.5% | +1.7       |        |               |
|        | 1000 w.              | 63.5% | +5.7       |        |               |
| Temp.  | 0 <sup>†</sup>       | 60.9% | —          | -.030  | < 0.001       |
|        | 0.25                 | 61.0% | +0.1       |        |               |
|        | 0.50                 | 60.7% | -0.1       |        |               |
|        | 0.75                 | 60.4% | -0.4       |        |               |
|        | 1.0                  | 60.5% | -0.3       |        |               |

allows stylistic variation ( $\sigma \in \{\text{low, moderate, high}\}$ ) but does *not* license factual content invention ( $\kappa = \text{low}$ ); any unsupported truth-conditional claim therefore counts as H.

### 4.1 PIPELINE OVERVIEW

Figure 1 summarizes the pipeline. We cross three task types, three creativity levels, and three target lengths over Wikipedia subjects to assemble 1,389 prompts, none granting permission to invent facts ( $\kappa$  low throughout) and generate 12,469 outputs from 10 models (Phases 1–2). Claims are extracted with **mimo-v2-flash** Xiaomi AI Lab [2026], verified against Wikipedia via dense retrieval and NLI following HalluLens [Bang et al., 2025] (Phase 3; 310K claims, 60% hallucinated on average), and analyzed (Phase 4; details in Appendix B).

### 4.2 EMPIRICAL FINDINGS

Table 2 consolidates the main effects and the task $\times$ creativity interaction. We highlight three qualitative phenomena central to our thesis:

**Writing format is the single largest driver.** The requested format dominates all other factors: LESSON\_PLAN produces 21% more hallucination than INTERVIEW ( $\rho = 0.45$ ), as visible in Figure 2 where the middle panel sits consistently above the other two regardless of creativity level or model.

**Creative intent raises hallucination monotonically.** Without permission to invent facts, moving from FACTUAL

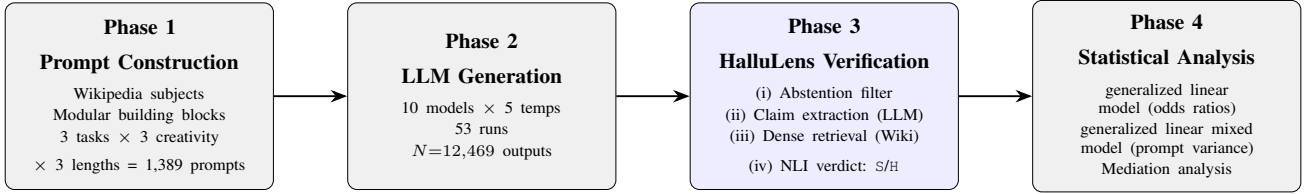


Figure 1: Four-phase experimental pipeline. Phases 1–2 construct and generate 12,469 outputs from 10 models by factorial crossing of task, creativity level, and length. Phase 3 applies the HalluLens verification paradigm [Bang et al., 2025]: atomic claim extraction, dense Wikipedia retrieval (gtr-t5-large), and NLI verdict (Supported / Hallucinated). Phase 4 analyses effects via GLM, GLMM, and mediation with false discovery rate (FDR) correction [Benjamini and Hochberg, 1995]. Full protocol in Appendix B.

to VERY\_CREATIVE raises hallucination by +12.8% (Table 2); every line in Figure 2 slopes upward across all three tasks. A mediation analysis confirms this is a direct effect on factual accuracy, not an artifact of longer or more claim-dense outputs. On a subset scored with CreativityPrism [Hou et al., 2025] ( $N=8,376$ ), divergent-thinking and writing-quality scores rise in lockstep with hallucination, confirming that the two are empirically coupled even under a regime that never licenses invention (full correlation matrices and claim-level distributions in Appendix A).

**Factual constraints moderate the creativity effect (TR-Relativity).** The task  $\times$  creativity interaction reveals a key asymmetry. LESSON\_PLAN and INTERVIEW show comparable creativity slopes (+10.6 and +14.3% swings respectively; interaction n.s.), differing only in baseline level. By contrast, NEWS\_ARTICLE shows a significantly dampened slope (+8.4%, interaction OR = 0.77,  $p_{\text{FDR}} < 10^{-77}$ ): journalism’s implicit factual norms constrain how much creative prompting inflates hallucination. This is visible in Figure 2, where NEWS\_ARTICLE slopes (right panel) are systematically flatter than those in INTERVIEW (left panel). The same creative instruction thus produces markedly different hallucination impacts depending on the format’s factual expectations—direct evidence for TR-relative hallucination.

Target output length has a significant but weaker effect ( $\rho = 0.11$ ,  $p < 10^{-32}$ ), roughly one-quarter the magnitude of the task effect; per-model profiles for all three tasks are shown in Figure 3 (Appendix D.1).

### 4.3 EVALUATOR RELIABILITY

Since our pipeline uses LLMs to judge hallucination, a natural concern is evaluator dependence [Bang et al., 2025, Dietz et al., 2025, Zheng et al., 2023]. We replicated the full pipeline with four judges (mimo-v2-flash (primary) Xiaomi AI Lab [2026], deepseek-v3.2 DeepSeek-AI [2025], llama-3.3-70b-instruct Meta AI [2024], and qwen3.5-397b-a17b Alibaba Cloud [2025]) on 90 shared prompts. All six evaluator pairs reach substantial agreement (mean  $\kappa_{\text{agree}} = 0.756$ , mean agreement = 88.1%; Table 11 in Appendix E). Absolu-

te hallucination rates shift by at most 6.4 % across judges, but the *relative ordering* of every experimental condition is preserved (Table 12). Thus, our conclusions are robust to evaluator choice.

## 5 RESEARCH AGENDA (OPEN PROBLEMS)

The framework and findings above motivate six open problems, organized from measurement foundations (items 1–3) to system-level design (items 4–6).

**Operationalizing CI and TR as scorable variables.** Our study operationalizes creative intent as a three-level categorical variable and truth regime as a fixed contract. For general-purpose evaluation, both constructs need finer-grained operationalizations. CI could be placed on a continuous scale via pairwise preference judgments fitted to a Bradley–Terry or Elo model, analogously to how arena-style leaderboards score overall quality [Boubdir et al.]. TR could be decomposed into its three components ( $O, \sigma, \kappa$ ) and each annotated independently, making regime boundaries explicit rather than implicit in the task label. A key challenge is inter-annotator reliability: creativity judgments are notoriously evaluator-dependent [Acar et al., 2019, Marco et al., 2025, Ismayilzada et al., 2025b], and regime boundaries involve normative choices (what *should* be grounded?) that require domain expertise.

**Automatic TR inference from prompts.** Users rarely specify a formal truth regime, so intent classifiers must infer  $\text{TR}(p)$  from prompt text [Wang et al., 2024b, Bodonhelyi et al., 2024]. Calibration is critical: misclassifying a strict-TR prompt as permissive suppresses warranted penalties, while the reverse penalizes legitimate PD. Hybrid approaches (classifier-based defaults with user override) and failure-mode analysis on ambiguous prompts deserve investigation.

**Metrics for “excessive” H under a given TR.** Current metrics treat hallucination rate as a single number. Under a TR-aware framework, what matters is *excess* deviation relative to the regime’s tolerance. This requires (i) sever-

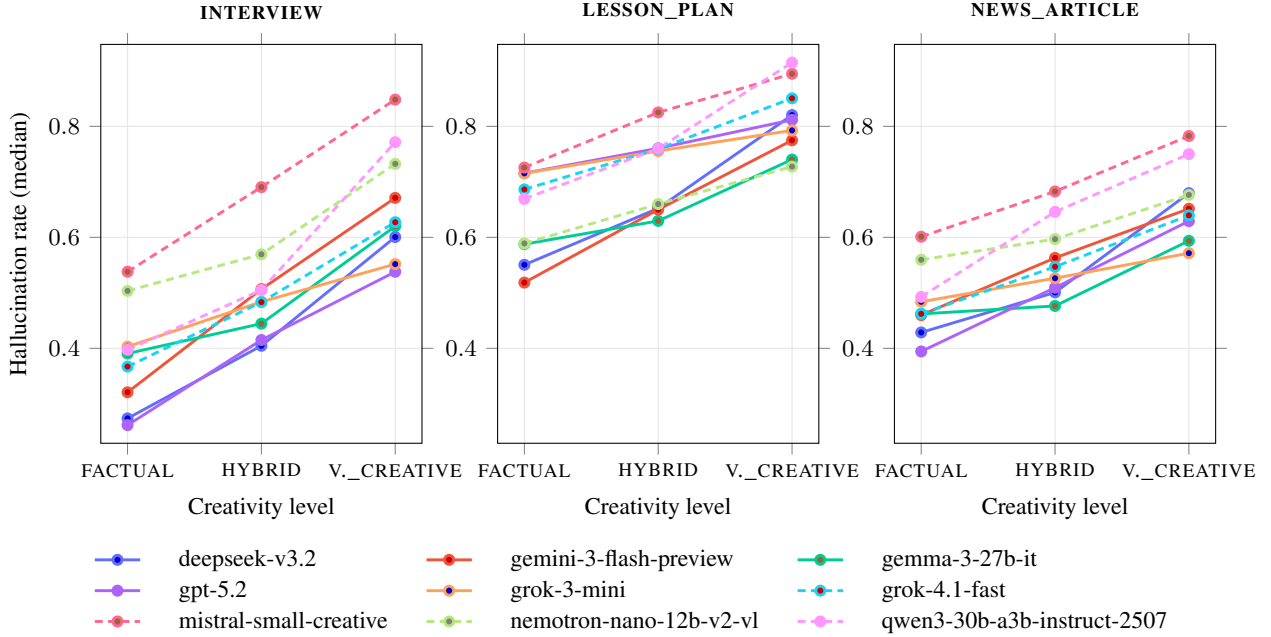


Figure 2: Median hallucination rate by model, task, and creativity level. Each panel shows one task; lines connect the same model across creativity levels. Two patterns emerge: (1) LESSON\_PLAN (middle) is consistently higher, and (2) upward slopes confirm the creativity effect, which is attenuated in NEWS\_ARTICLE (right). Verification follows HalluLens [Bang et al., 2025].

ity weighting (a fabricated safety claim is worse than a minor date error), (ii) claim-level vs. output-level aggregation (a single critical hallucination may outweigh many benign ones), and (iii) regime-conditioned baselines (the “acceptable” rate differs between grounded summarization and brainstorming). Existing severity-aware proposals in hallucination evaluation [Bang et al., 2025, Huang et al., 2025] and risk-weighted approaches in factuality benchmarks [Wei et al., 2024, Lin et al., 2022] provide starting points, but none currently conditions severity on an explicit TR contract.

**Robust H detection under stylistic variation.** Because stylistic creativity alone raises measured hallucination, NLI-based verifiers [Aynedtinov and Akbik, 2024, Chen and Eger, 2023] may conflate figurative language with unsupported claims. A dedicated suite pairing identical factual content with varying stylistic treatments would isolate verifier brittleness from genuine divergence [Jin et al., 2022].

**Controlling H without collapsing PD.** Mitigation techniques (constrained decoding, retrieval augmentation, post-hoc verification) typically target H reduction without measuring the collateral effect on creative quality. Recent work shows these trade-offs are asymmetric: verification style interventions can preserve divergent thinking, while decoding constraints may suppress it [Banerjee et al., 2025, He et al., 2025a, Sun et al., 2024, Li et al.]. TR-aware evaluation should therefore report *both* H reduction and PD preservation under the same regime.

**Benchmarks crossing task  $\times$  CI  $\times$  TR.** No existing benchmark systematically crosses task type, creative intent level, and truth regime as independent variables. Such a benchmark needs a task gradient (summarization  $\rightarrow$  style transfer  $\rightarrow$  fiction), multi-level CI manipulations, explicit ( $O, \sigma, \kappa$ ) annotations, and protocols scoring H and PD separately. CreativityPrism [Hou et al., 2025] and LitBench [Fein et al., 2025] provide partial coverage but neither conditions scores on an explicit TR.

## 6 DISCUSSION

### 6.1 IMPLICATIONS FOR EVALUATION PRACTICE

The practical consequence of TR-relative evaluation is a shift from defect counting to *regime-conditioned diagnosis*. Rather than asking “how much does the output hallucinate?”, evaluation should ask “does its divergence exceed what the TR licenses?” Three verdicts result: *aligned* (divergence matches TR), *excessive-H* (unsupported claims where grounding is required), or *insufficient-PD* (output too conservative for a permissive TR). Because writing format alone shifts measured hallucination by over 20 % (Table 2), regime-specific baselines are essential: identical rates carry different meanings under strict and permissive contracts. At the same time, evaluation under permissive regimes should still flag *truth-commitment violations*—instances where the



model presents invented content as factual rather than as hypothesis or creative proposal [Bang et al., 2025, Hou et al., 2025].

## 6.2 WHAT DRIVES MEASURED HALLUCINATION

Three findings refine the picture of what actually drives measured H.

**The prompt is part of the hallucination.** Prompt-level variance is substantial (intraclass correlation coefficient (ICC) = 0.65): the same model at the same temperature exhibits markedly different H rates depending on prompt content and structure. H is therefore not purely a model deficiency; prompt engineering matters as much as model selection, and benchmarks should sample prompts diversely enough to capture this variance.

**Temperature is not the main control knob.** We find no meaningful association between temperature and H ( $r = -0.006$ ,  $p = 0.52$ ), a null result robust across correlation methods, GLM, and mixed models. This aligns with recent evidence that temperature is a weak CI parameter [Du et al., 2025, Peeperkorn et al., 2024]: semantic-level signals in the prompt appear to dominate token-level sampling noise.

**Format-level factual constraints moderate the creativity effect.** The interaction between CI level and task type is significant (OR = 0.77 for VERY\_CREATIVE  $\times$  NEWS\_ARTICLE): when a writing format carries strong factual expectations, creative prompting has a weaker effect on H, suggesting the model partially internalizes those constraints. This mirrors the layer-wise CI/H trade-offs reported by He et al. [2025a], though our evidence arises from prompt-level manipulations rather than model-internal dynamics. Together, these three patterns reinforce recent critiques of task-agnostic evaluation [Bang et al., 2025, Wang et al., 2024b, Cyriac, 2025] and identify prompt content, not sampling temperature, as the first-order confound that benchmarks must control.

## 6.3 LIMITATIONS

Three design choices bound the scope of our study without undermining its core conclusions.

**Binary verification.** Our claim-level pipeline uses supported/hallucinated verdicts, following the state-of-the-art HalluLens protocol [Bang et al., 2025], and omits an “unverifiable” category. This may overestimate absolute hallucination rates under creative conditions, where invented but unfalsifiable content is common; however, because the same classification applies uniformly across all

conditions, the *relative* comparisons that support our claims (effect of CI, task type, and their interaction) remain valid.

**Knowledge-base scope.** Wikipedia serves as the verification oracle, which is standard practice in factuality evaluation [Thorne et al., 2018, Lin et al., 2022]. Complementary sources (domain-specific corpora, recent-event databases) could improve coverage for specialized or time-sensitive claims, but Wikipedia’s breadth and neutrality make it a reasonable default for the general-knowledge prompts used here.

**Prompt and TR alignment.** Some prompts in our design are ambiguous with respect to the intended truth regime: open-ended formulations can make it difficult to determine whether a given deviation is TR-licensed or not. Extending the task gradient to genres with sharper regime boundaries (e.g., fiction vs. legal analysis) and annotating each prompt with an explicit ( $O, \sigma, \kappa$ ) contract would strengthen future replications.

## 7 CONCLUSION

Our controlled study (12,469 texts, 10 models, 310K claims) shows that creative intent, writing format, and format-specific factual constraints jointly shape hallucination rates in a TR-relative manner. The minimal vocabulary introduced here (TR, CI, PD, H) makes the prompt contract explicit and turns evaluation from binary defect counting into regime-conditioned diagnosis: does the output’s divergence exceed, match, or fall short of what its TR licenses? The six open problems outlined in Section 5 chart a path toward evaluation infrastructure that can penalize harmful fabrication under strict regimes while preserving productive divergence where exploration is intended.

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## A EXTENDED STATISTICAL RESULTS

### A.1 FULL CORRELATION MATRIX

Table 3 presents robust Pearson and Spearman correlations between all continuous experimental factors and the prompt-level hallucination rate.

### A.2 SPEARMAN CORRELATIONS BY FACTOR (FULL RANKING)

Table 4 ranks *all* experimental factors, including categorical one-hot encodings and individual model names, by  $|\rho|$  with the hallucination rate.

### A.3 PARTIAL CORRELATIONS

Table 5 shows correlations controlling for response length and claim count, isolating each factor’s *direct* association with hallucination.

### A.4 CLAIM-LEVEL VERIFICATION DISTRIBUTION

Table 6 reports the raw verdict counts from the NLI pipeline before prompt-level aggregation.

Among verified claims only (excluding “no claim”), the support rate is 41.7% and the hallucination rate is 58.3%.

### A.5 CLAIM-LEVEL SUPPORT RATE BY TASK $\times$ CREATIVITY

Table 7 disaggregates claim-level support rates across the full  $3 \times 3$  task-by-creativity design.

The claim-level results confirm the prompt-level patterns from Table 2: lesson plans have the lowest support rate in every creativity condition, and support rates decrease monotonically with increasing creative intent across all three task types.

### A.6 CREATIVITY-HALLUCINATION CORRELATION MATRIX

Table 8 presents the full Spearman correlation matrix between creativity scores and hallucination metrics for the 29 runs with CreativityPrism data ( $N = 8,376$ ).

Hallucination rate correlates with TTCT (divergent thinking,  $\rho = 0.23$ ) and with the creativity composite ( $\rho = 0.17$ ), while the correlation with TTCW (writing quality) is negligible ( $\rho \approx 0$ ). This pattern is consistent with the  $\sigma/\kappa$  distinction: stylistic latitude ( $\sigma$ ) alone increases measured divergence from factual baselines, even when the prompt never licenses factual invention ( $\kappa$  low).

## B EXPERIMENTAL DETAILS

### B.1 FORMAL CLAIM-LEVEL DECISION RULE

Given prompt  $p$  and output  $y$ :

Table 3: Robust correlations with hallucination rate ( $N = 12,469$ ).

| Metric                 | Method   | $r$    | 95% CI          | $p$         | $pFDR$      |
|------------------------|----------|--------|-----------------|-------------|-------------|
| creativity_rank        | Pearson  | 0.215  | [0.197, 0.233]  | $< 10^{-4}$ | $< 10^{-4}$ |
| length_words           | Pearson  | 0.155  | [0.136, 0.174]  | $< 10^{-4}$ | $< 10^{-4}$ |
| response_length_tokens | Pearson  | 0.126  | [0.107, 0.145]  | $< 10^{-4}$ | $< 10^{-4}$ |
| response_length_words  | Pearson  | 0.113  | [0.095, 0.132]  | $< 10^{-4}$ | $< 10^{-4}$ |
| n_claim_rows           | Pearson  | 0.070  | [0.050, 0.089]  | $< 10^{-4}$ | $< 10^{-4}$ |
| temperature            | Pearson  | -0.006 | [-0.026, 0.012] | 0.525       | 0.537       |
| creativity_rank        | Spearman | 0.215  | [0.197, 0.233]  | $< 10^{-4}$ | $< 10^{-4}$ |
| length_words           | Spearman | 0.153  | [0.133, 0.173]  | $< 10^{-4}$ | $< 10^{-4}$ |
| temperature            | Spearman | -0.006 | [-0.025, 0.013] | 0.537       | 0.537       |

1. Infer or specify the regime  $TR(p)$ : what is truth-conditional, what is evidence-bound, what is optionally speculative, and what (if anything) may be invented [Bang et al., 2025, Huang et al., 2025].
2. Decompose  $y$  into atomic claims at the chosen granularity (ideally claim-level) [Huang et al., 2025, Ji et al., 2023].
3. For each claim  $c$ :
  - if  $c$  is truth-conditional under  $TR(p)$  and unsupported/incorrect relative to the oracle, label it H [Ji et al., 2023, Huang et al., 2025];
  - else if  $c$  is permitted under  $TR(p)$  and improves usefulness/appropriateness while remaining constraint-compliant, label it PD [Acar et al., 2019, Runco and Jaeger, 2012];
  - else label it *neither* (e.g., purely stylistic variation or benign noise).

## B.2 PROMPT TEMPLATE STRUCTURE

See Appendix F for full block definitions. An `AblationConfig` dataclass allows toggling each block independently, enabling future ablation studies.

This modular design ensures full factorial crossing: 3 tasks  $\times$  3 creativity levels  $\times$  3 lengths  $\times$   $k$  subjects = 1,389 unique prompts.

## B.3 FACTHALU VERIFICATION PIPELINE

The claim-level verification pipeline follows the HalluLens evaluation paradigm [Bang et al., 2025] and proceeds in four stages:

1. **Abstention filter.** An LLM judges whether the generated text constitutes a genuine attempt at the task or an explicit refusal/hedge. Abstained outputs are excluded from analysis.

2. **Claim extraction.** A prompted LLM extracts atomic, independently verifiable claims from each sentence. Context is provided via a sliding window with `<SOS>`/`<EOS>` markers to preserve discourse-level meaning.
3. **Evidence retrieval.** For each extracted claim, we perform hybrid retrieval against the English Wikipedia (April 2023 dump): BM25 (lexical) and dense retrieval (`gtr-t5-large` encoder). The top-5 passages per claim are concatenated as evidence context.
4. **NLI verdict.** A verifier LLM issues a binary verdict (supported or hallucinated) given the claim, the evidence passages, and the original generation context. The primary evaluator for all three LLM stages is **mimo-v2-flash** Xiaomi AI Lab [2026].

To assess evaluator reliability, we replicate the full pipeline on 90 shared prompts ( $n = 1,943$  claims) with four different evaluator LLMs: **mimo-v2-flash** Xiaomi AI Lab [2026], **deepseek-v3.2** DeepSeek-AI [2025], **llama-3.3-70b-instruct** Meta AI [2024], and **qwen3.5-397b-a17b** Alibaba Cloud [2025].

**Prompt-level aggregation.** We compute  $\text{hallucination\_rate} = n_{\text{hallucinated}}/n_{\text{verified}}$  at the prompt level, excluding `no_claim` rows. The binary scheme (no “unverifiable” category) likely overestimates absolute rates but preserves relative ordering across conditions (Section 6.3).

## B.4 RUN COVERAGE SUMMARY

The study comprises 53 experimental runs spanning 10 models across five providers: **deepseek-v3.2** DeepSeek-AI [2025], **gemini-3-flash-preview** Google DeepMind [2025a], **gemma-3-27b-it** Google DeepMind [2025b], **gpt-5.2** OpenAI [2025a,b], **grok-3-mini xAI** [2025a], **grok-4.1-fast xAI** [2025b], **mistral-small-creative** Mistral AI [2025], **nemotron-nano-12b-v2-v1** NVIDIA [2025], and **qwen3-30b-a3b-instruct-2507** Alibaba Cloud [2025], with tem-



Table 4: Spearman correlations by factor (top 20 effects,  $N = 12,469$ ). Encoding: one-hot for categorical levels; raw ranks for numeric factors.

| Factor                | Modality             | $\rho$ | $ \rho $ | $p$           | Strength  |
|-----------------------|----------------------|--------|----------|---------------|-----------|
| task                  | LESSON_PLAN          | 0.450  | 0.450    | $< 10^{-300}$ | moderate  |
| task                  | INTERVIEW            | -0.301 | 0.301    | $< 10^{-258}$ | moderate  |
| creativity_rank       | (global)             | 0.295  | 0.295    | $< 10^{-249}$ | weak      |
| creativity_level      | VERY_CREATIVE        | 0.263  | 0.263    | $< 10^{-195}$ | weak      |
| creativity_level      | FACTUAL              | -0.249 | 0.249    | $< 10^{-174}$ | weak      |
| task                  | NEWS_ARTICLE         | -0.151 | 0.151    | $< 10^{-64}$  | weak      |
| model_name            | mistral-sm.-creative | 0.137  | 0.137    | $< 10^{-53}$  | weak      |
| length_words          | (global)             | 0.106  | 0.106    | $< 10^{-32}$  | weak      |
| resp. length (tok.)   | (global)             | 0.085  | 0.085    | $< 10^{-21}$  | very weak |
| model_name            | gemini-3-flash       | -0.083 | 0.083    | $< 10^{-20}$  | very weak |
| model_name            | deepseek-v3.2        | -0.062 | 0.062    | $< 10^{-11}$  | very weak |
| model_name            | qwen3-30b-a3b        | 0.060  | 0.060    | $< 10^{-11}$  | very weak |
| resp. length (words)  | (global)             | 0.053  | 0.053    | $< 10^{-8}$   | very weak |
| model_name            | gemma-3-27b-it       | -0.048 | 0.048    | $< 10^{-7}$   | very weak |
| model_name            | gpt-5.2              | -0.047 | 0.047    | $< 10^{-7}$   | very weak |
| prompt length (words) | (global)             | -0.041 | 0.041    | $< 10^{-5}$   | very weak |
| model_name            | grok-3-mini          | 0.035  | 0.035    | $< 10^{-4}$   | very weak |
| temperature           | (global)             | -0.030 | 0.030    | 0.001         | very weak |
| model_name            | nemotron-nano-12b    | 0.027  | 0.027    | 0.002         | very weak |
| n_claims              | (global)             | 0.024  | 0.024    | 0.007         | very weak |
| creativity_level      | HYBRID               | -0.013 | 0.013    | 0.140         | n.s.      |
| model_name            | grok-4.1-fast        | -0.003 | 0.003    | 0.763         | n.s.      |

Table 5: Partial correlations (controlling for response length and claim count).

| Metric          | $r_{\text{partial}}$ | 95% CI           | $p$          | $p_{\text{FDR}}$ |
|-----------------|----------------------|------------------|--------------|------------------|
| creativity_rank | 0.195                | [0.177, 0.213]   | $< 10^{-91}$ | $< 10^{-91}$     |
| length_words    | 0.176                | [0.158, 0.195]   | $< 10^{-75}$ | $< 10^{-74}$     |
| temperature     | -0.022               | [-0.042, -0.003] | 0.023        | 0.023            |

Table 6: Overall claim-level verdict distribution ( $N_{\text{claims}} = 310,357$ ).

| Verdict             | Count          | %           |
|---------------------|----------------|-------------|
| Hallucinated        | 157,761        | 50.8%       |
| Supported           | 113,046        | 36.4%       |
| No claim / filtered | 39,550         | 12.7%       |
| <b>Total</b>        | <b>310,357</b> | <b>100%</b> |

perature  $\in \{0, 0.25, 0.5, 0.75, 1.0\}$  and 3 target lengths. CreativityPrism scoring is available for 29 complete runs ( $N = 8,376$ ); the remaining 24 runs lack creativity annotations due to pipeline coverage gaps. Table 9 summarizes coverage by model.

## B.5 STATISTICAL ANALYSIS

To ensure our findings are robust, we use several statistical methods. First, we use logistic regression (a binomial GLM with a logit link) to measure how much each factor (task, creativity, length) multiplies the odds of a hallucination occurring (reported as Odds Ratios, OR) [Nelder and Wedderburn, 1972, Hosmer Jr et al., 2013]. Second, we use a generalized linear mixed model (GLMM) to calculate how much of the hallucination rate is driven by the specific prompt topic itself, rather than our experimental settings [Breslow and Clayton, 1993, Bates et al., 2015]. Third, we use mediation analysis to verify that creative prompts cause hallucinations directly, rather than simply causing the model to write more claims [Baron and Kenny, 1986, Imai et al., 2010]. Finally, to prevent false positives from running multiple tests, all  $p$ -values are adjusted using FDR correction [Benjamini and Hochberg, 1995, Benjamini and Yekutieli, 2001], and correlations are verified through 5,000 bootstrap iterations [Efron and Tibshirani, 1993].

Table 7: Claim-level support rate by task and creativity level (verified claims only).

| Task         | Creativity    | $n_{\text{verified}}$ | $n_{\text{supported}}$ | Support rate |
|--------------|---------------|-----------------------|------------------------|--------------|
| INTERVIEW    | FACTUAL       | 37,408                | 21,106                 | 56.4%        |
| INTERVIEW    | HYBRID        | 40,293                | 20,233                 | 50.2%        |
| INTERVIEW    | VERY_CREATIVE | 36,314                | 16,147                 | 44.5%        |
| LESSON_PLAN  | FACTUAL       | 22,352                | 6,376                  | 28.5%        |
| LESSON_PLAN  | HYBRID        | 24,082                | 6,027                  | 25.0%        |
| LESSON_PLAN  | VERY_CREATIVE | 24,643                | 5,064                  | 20.5%        |
| NEWS_ARTICLE | FACTUAL       | 29,108                | 13,696                 | 47.1%        |
| NEWS_ARTICLE | HYBRID        | 29,524                | 13,269                 | 44.9%        |
| NEWS_ARTICLE | VERY_CREATIVE | 27,083                | 11,128                 | 41.1%        |

Table 8: Spearman correlation matrix: creativity and hallucination metrics ( $N = 8,376$ , 29 runs). TTCT = divergent thinking score; TTCW = creative writing quality; composite = weighted combination.

|              | hallu. rate | n_claims | resp. length | TTCT  | TTCW   | composite |
|--------------|-------------|----------|--------------|-------|--------|-----------|
| hallu. rate  | 1.000       | −0.008   | 0.032        | 0.228 | −0.004 | 0.169     |
| n_claims     | −0.008      | 1.000    | 0.874        | 0.260 | 0.106  | 0.257     |
| resp. length | 0.032       | 0.874    | 1.000        | 0.370 | 0.148  | 0.360     |
| TTCT         | 0.228       | 0.260    | 0.370        | 1.000 | 0.260  | 0.888     |
| TTCW         | −0.004      | 0.106    | 0.148        | 0.260 | 1.000  | 0.642     |
| composite    | 0.169       | 0.257    | 0.360        | 0.888 | 0.642  | 1.000     |

Table 9: Run coverage by model (53 total runs; 29 with creativity scores).

| Model                   | Total runs | w/ creativity |
|-------------------------|------------|---------------|
| grok-3-mini             | 13         | 12            |
| gemini-3-flash-preview  | 10         | 2             |
| deepseek-v3.2           | 3          | 3             |
| gemma-3-27b-it          | 3          | 3             |
| gpt-5.2                 | 6          | 0             |
| grok-4.1-fast           | 6          | 1             |
| mistral-small-creative  | 3          | 3             |
| nemotron-nano-12b-v2-vl | 3          | 3             |
| qwen3-30b-a3b-instruct  | 6          | 2             |

Table 10: Creativity and hallucination by task (CreativityPrism subset,  $N = 8,376$ ).

| Task         | $n$   | Hallu. rate | TTCT | TTCW |
|--------------|-------|-------------|------|------|
| INTERVIEW    | 2,764 | 52.3%       | 4.25 | 1.19 |
| LESSON_PLAN  | 2,813 | 73.6%       | 4.22 | 1.02 |
| NEWS_ARTICLE | 2,799 | 56.6%       | 4.15 | 1.58 |

## C ADDITIONAL TABLES

### C.1 CREATIVITY METRICS BY TASK

Table 10 breaks down creativity and hallucination scores by task type, showing that LESSON\_PLAN has the highest hallucination rate but *not* the highest creativity scores.

This dissociation (lesson plans hallucinate most but do not score highest on any creativity dimension) supports the thesis that H is driven by task constraints (TR) rather than by creativity *per se*.

## D ADDITIONAL FIGURES AND DETAILED ANALYSIS

### D.1 LENGTH EFFECT: HALLUCINATION RATE BY OUTPUT LENGTH

**Reading Figure 3.** The length effect is visible but modest. In the INTERVIEW panel (left), most models show a gentle upward trend from 250 to 1,000 words, with the exception of `deepseek-v3.2` (blue), which remains nearly flat. The LESSON\_PLAN panel (middle) is again the highest overall, but slopes are compressed: models that already hallucinate heavily at 250 words have little room to increase further (a possible ceiling effect). In the NEWS\_ARTICLE panel (right), some models (e.g., `grok-4.1-fast`, green) show a steeper rise, while others remain stable. Overall, the length effect ( $\rho = 0.11$ ) is roughly one-third the magnitude of the creativity effect ( $\rho = 0.30$ ) and one-quarter of the task effect ( $\rho = 0.45$ ), reinforcing that *what* the model is asked to do matters far more than *how long* it writes.

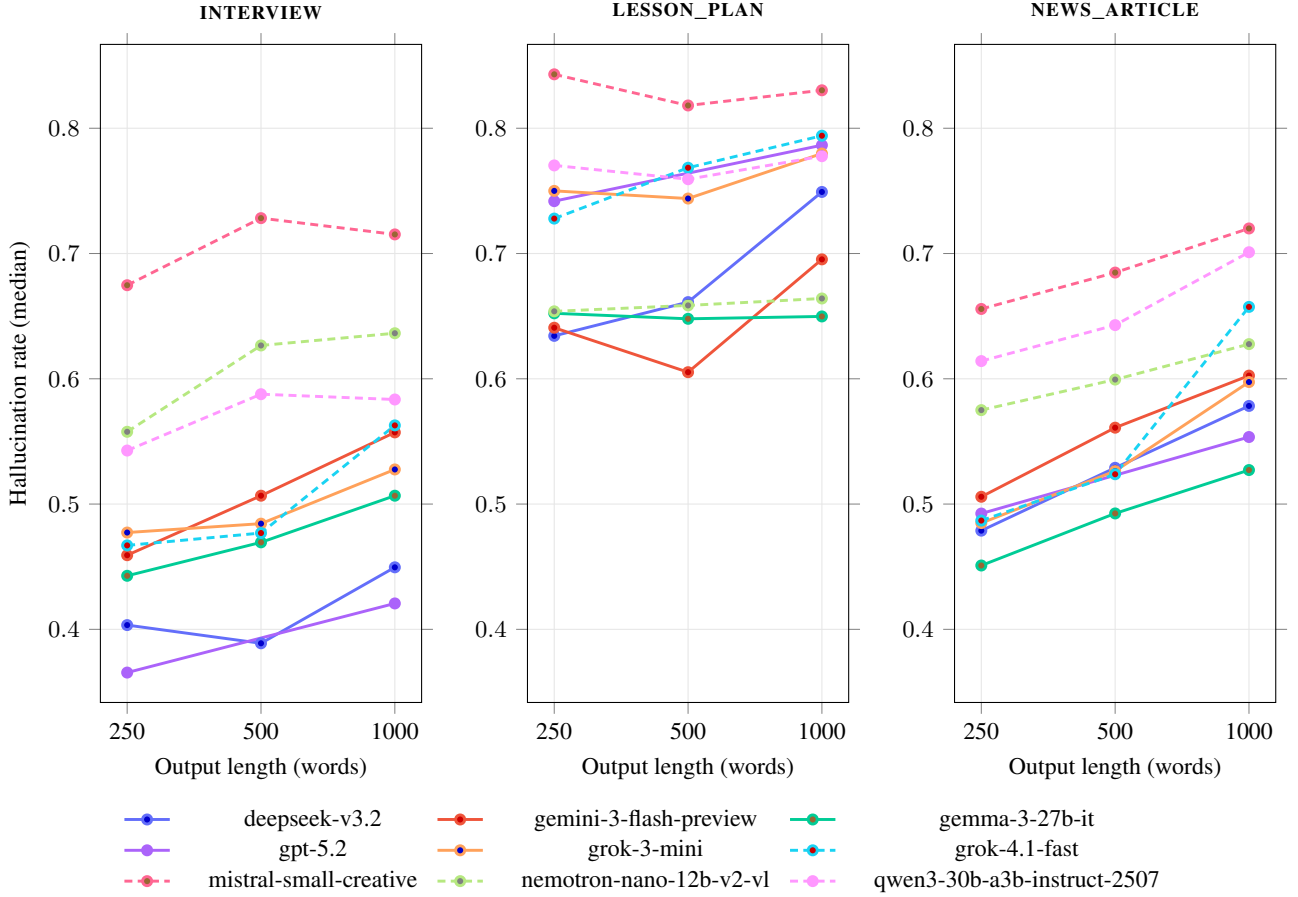


Figure 3: Median hallucination rate by model, task, and requested length. Each panel shows one task; lines connect the same model across output lengths. Longer outputs show moderately higher hallucination rates, but the effect is much weaker than that of task type or creativity level. Verification follows HalluLens [Bang et al., 2025].

## E EVALUATOR RELIABILITY: FULL TABLES

Table 11: Pairwise inter-evaluator agreement ( $n = 1,943$  comparable claims). All six pairs reach “substantial” agreement ( $\kappa_{\text{agree}} > 0.7$ ).

| Evaluator pair          | $n$ | Agree.       | $\kappa_{\text{agree}}$ |
|-------------------------|-----|--------------|-------------------------|
| llama-3.3 vs. mimo-v2   | 943 | 88.9%        | 0.778                   |
| mimo-v2 vs. qwen3.5     | 459 | 88.9%        | 0.768                   |
| deepseek vs. qwen3.5    | 395 | 88.4%        | 0.763                   |
| deepseek vs. mimo-v2    | 708 | 87.9%        | 0.755                   |
| deepseek vs. llama-3.3  | 682 | 87.5%        | 0.746                   |
| llama-3.3 vs. qwen3.5   | 467 | 86.9%        | 0.723                   |
| <b>Mean (all pairs)</b> | —   | <b>88.1%</b> | <b>0.756</b>            |

Table 12: Hallucination rate by evaluator LLM (same 90 prompts). The maximum absolute gap is 6.4 %; the relative ordering of conditions is identical across all four judges.

| Evaluator               | $n$ claims | Hallu. rate  |
|-------------------------|------------|--------------|
| mimo-v2-flash (primary) | 4,748      | 56.6%        |
| deepseek-v3.2           | 3,384      | 52.5%        |
| llama-3.3-70b-instruct  | 4,820      | 50.7%        |
| qwen3.5-397b-a17b       | 1,569      | 50.2%        |
| <b>Max gap</b>          |            | <b>6.4 %</b> |

## F PROMPT LEGO BLOCKS: FULL BLOCK DEFINITIONS

All blocks are defined in `utils/prompt_lego_blocks.py`. The module is structured as six independent components (A–F) that `STATIC_USER_PROMPT_TEMPLATE` assembles deterministically.

## F.1 BLOCK A — CREATIVITY STYLE (CREATIVITY\_STYLE)

**FACTUAL** “Tone: neutral, informative, straightforward. Avoid metaphors, theatrics, and flowery language. Focus on clarity and precision.”

**HYBRID** “Tone: clear and engaging with light creative flourishes (rhythm, mild imagery, occasional vivid word choice), staying focused on the topic. Balance accessibility with style.”

**VERY\_CREATIVE** “Tone: highly stylized and distinctive (strong voice, playful constraints, vivid imagery, unexpected angles), but still centered on the topic. The form itself becomes part of the content.”

## F.2 BLOCK B — TASK CONSTRAINTS (TASK\_CONSTRAINTS), SUBSET USED

**INTERVIEW** *Use-case: Q&A for a community newsletter; accessible, skimmable.*

- Exactly 8 Q/A pairs; markers Q1:/A1: ... Q8:/A8:.
- Answers: 2–4 sentences each.
- Questions progress from basic to nuanced.

**NEWS\_ARTICLE** *Use-case: news-style explainer for a general audience on deadline.*

- Headline (one line, compelling).
- Deck: exactly 2 sentences.
- 3 sections with clear subheadings.
- Exactly 2 direct quotes in double quotes (attributed to named sources).
- 1 forward-looking perspective sentence at the end.

**LESSON\_PLAN** *Use-case: teacher lesson plan for a class.*

- Grade level specified at top.
- Learning objectives: exactly 3.
- Materials needed: list format.
- 1 activity with time allocations.
- Assessment criteria section.

Task constraints are *identical across all creativity levels*: Block A (style) and Block B (format) are orthogonal by design. The full TaskName enum lists 16 task types; only these 3 are used in our study.

## F.3 BLOCK C — LENGTH (LENGTH\_BLOCK)

Target length: {length\_words} words  
(plus or minus 15%).

We use  $\lambda \in \{250, 500, 1,000\}$ .

## F.4 BLOCK D — BANNED PHRASES (BANNED\_PHRASES)

The following 19 terms are forbidden in generated outputs to prevent benchmark contamination and suppress LLM meta-commentary about its own uncertainty:

*wikipedia, wikidata, benchmark, evaluation, oracle, rag, retrieval, closed-book, hallucination, ground truth, cite sources, citations required, as an ai, system prompt, language model, trained on, knowledge cutoff, i don't have access, i cannot browse.*

## F.5 BLOCKS E–F — PROMPT SCHEMAS AND FINAL TEMPLATE

Block E (BLUEPRINT\_SCHEMA) and Block F (FINAL\_PROMPT\_SCHEMA) support a two-stage LLM generation path (blueprint JSON  $\rightarrow$  naturalized user message). For reproducibility, all 1,389 prompts use the deterministic STATIC\_USER\_PROMPT\_TEMPLATE instead, whose OUTPUT\_RULES block encodes the strict TR at every creativity level:

*“If you’re unsure about a specific detail, keep it high-level or phrase it cautiously rather than guessing.”*

Together with BANNED\_PHRASES, this instruction prevents models from filling knowledge gaps with confidently stated inventions, bounding the confounding between  $\sigma$  (style latitude) and  $\kappa$  (invention license).